Avishan Ketheswaran

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Education

Toronto Metropolitan University (TMU) — *Bachelor of Mechatronics Engineering* Sept 2025 – Present

Experience

TMU Hyperloop Design team — *structural team member* *Sept 2025 – Present*

- Designed and CAD-modeled the pod aeroshell in SolidWorks using a multi-plane loft workflow, targeting aerodynamic efficiency for wind tunnel testing
- Ran ANSYS FEA on structural components to validate load paths, safety factors, and mounting feasibility
- Developed a magnetic braking system mounting concept balancing structural constraints, safety factors, and manufacturability

Geek Squad (Best buy Canada) — *Junior Agent* *June 2026 – Present*

- Diagnosed and repaired hardware and software faults on laptops, desktops, and mobile devices in a high-volume downtown location
- Prioritized repair queues with structured triage; documented intake and diagnostics to support warranty tracking
- Explained technical issues and repair status to customers in plain language, supporting first-visit resolutions

Micro Kool Computers — *Computer Repair Technician Intern* *Jan 2024 – June 2024*

- Cut repair turnaround time by ~25% through triage-based prioritization of high-volume daily intake
- Completed 15+ device repairs per week across PCs, laptops, and peripherals alongside senior technicians
- Handled front-desk intake, parts sourcing, and invoicing

Mechatronics Course Union — Director of Events *Sept 2025 – Present*

- Supported end-to-end execution of a flagship technical student event as part of the events team
- Assembled and finished a full wooden competition arena using cutting tools and fabrication methods
- Resolved issues in real time and kept all competing teams aligned and ready throughout the event

Projects / Co-Curriculars

MARS Rocketry Competition — Team Lead, Team Icarus

- Led a 5-member team through design, construction, and launch of a competition rocket, delivered on schedule
- Modeled stability margins, motor selection, and predicted apogee in OpenRocket before the build
- Launched on a G-class motor to an apogee of ~550 ft

Custom Arcade Machine — Independent Engineering Project

- Designed and built a full mechatronic system: structural enclosure, embedded electronics, and a Java-based software platform
- Wrote modular OOP software for game logic, controller mapping, and hardware abstraction
- Debugged signal, power, and peripheral compatibility issues through iterative hardware-software integration testing.

Skills

- **CAD / Simulation:** SolidWorks (3D modeling, assemblies, lofts), ANSYS (FEA, structural analysis),
- **Programming:** C, Python (scripting, automation), Java (OOP), MATLAB (simulation, data analysis)
- **Machining**: Miter Saw, Jig Saw, Circular Saw, Belt Sander, Woodworking, 3D Printing, 2D drawings
- **Hardware:** PC assembly & diagnostics, circuit troubleshooting, electronic system integration
- **Software:** Microsoft Office Suite (Excel advanced formulas, Word, PowerPoint, Teams), Git